

OptiSight - Lightweight Vision Model Comparison

Weight Class	Model Family	Full Model Name	Parameters	Why Suitable for OptiSight?	Weakness
Ultra Lightweight (Speed Priority)	Moondream	Moondream2	1.9B	Speed demon. Analyzes 10-15 frames/sec. Perfect for sudden obstacle detection.	Cannot build complex logic (CoT). Says "stone exists" but may skip "turn left because stone."
Ultra Lightweight (Code-First)	DeepSeek	DeepSeek-VL-1.3B	1.3B	Code and logic focused. Strictly follows JSON formats like {"action": "turn_left"}	Visual perception may not be as sharp as Qwen.
Lightweight (Balanced)	InternVL	InternVL2-1B / 2B	1B / 2B	Highly detailed. Best at explaining "Why did I stop?" Describes scene objects thoroughly.	Sometimes prefers narrating over giving commands (fixable with prompting).
Lightweight (★ Best Choice)	Qwen	Qwen2-VL-2B-Instruct	2.2B	Best CoT and logic. Supports video and high resolution. Excellent at recognizing agricultural objects (plants, stones, tractors).	Inference speed slightly slower than 1B models.
Compact (OCR Specialist)	MiniCPM	MiniCPM-V 2.6	2-3B	Powerful OCR and detail detection. Can spot small objects (e.g., plant diseases).	Slightly heavier model.

Note: ★ Qwen2-VL-2B-Instruct is recommended as the primary model for OptiSight due to its optimal balance of speed, accuracy, and agricultural object recognition.